

Tables, Graphs, and Descriptive Statistics

QBA 5330 — Module 1, Part 3 (September 5)

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We know how to frame a decision, and where trustworthy data comes from.

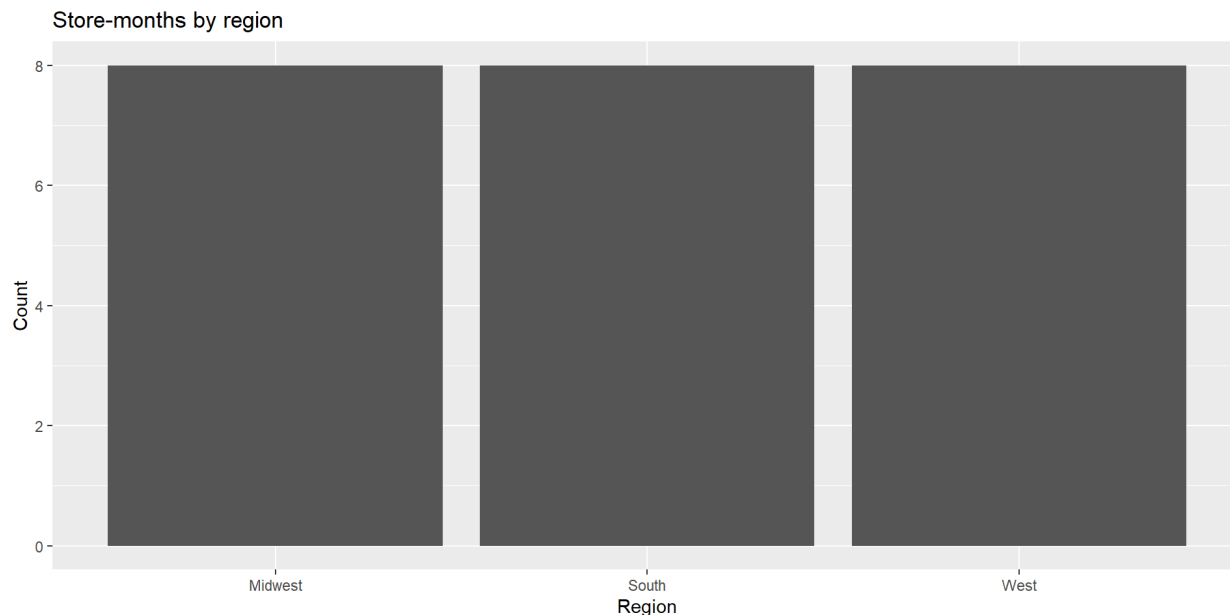
Now: how do we actually look at it?

Matching the Graph to the Variable

- **Categorical variable** → frequency table → **bar chart**
- **Quantitative variable** → binned counts → **histogram**

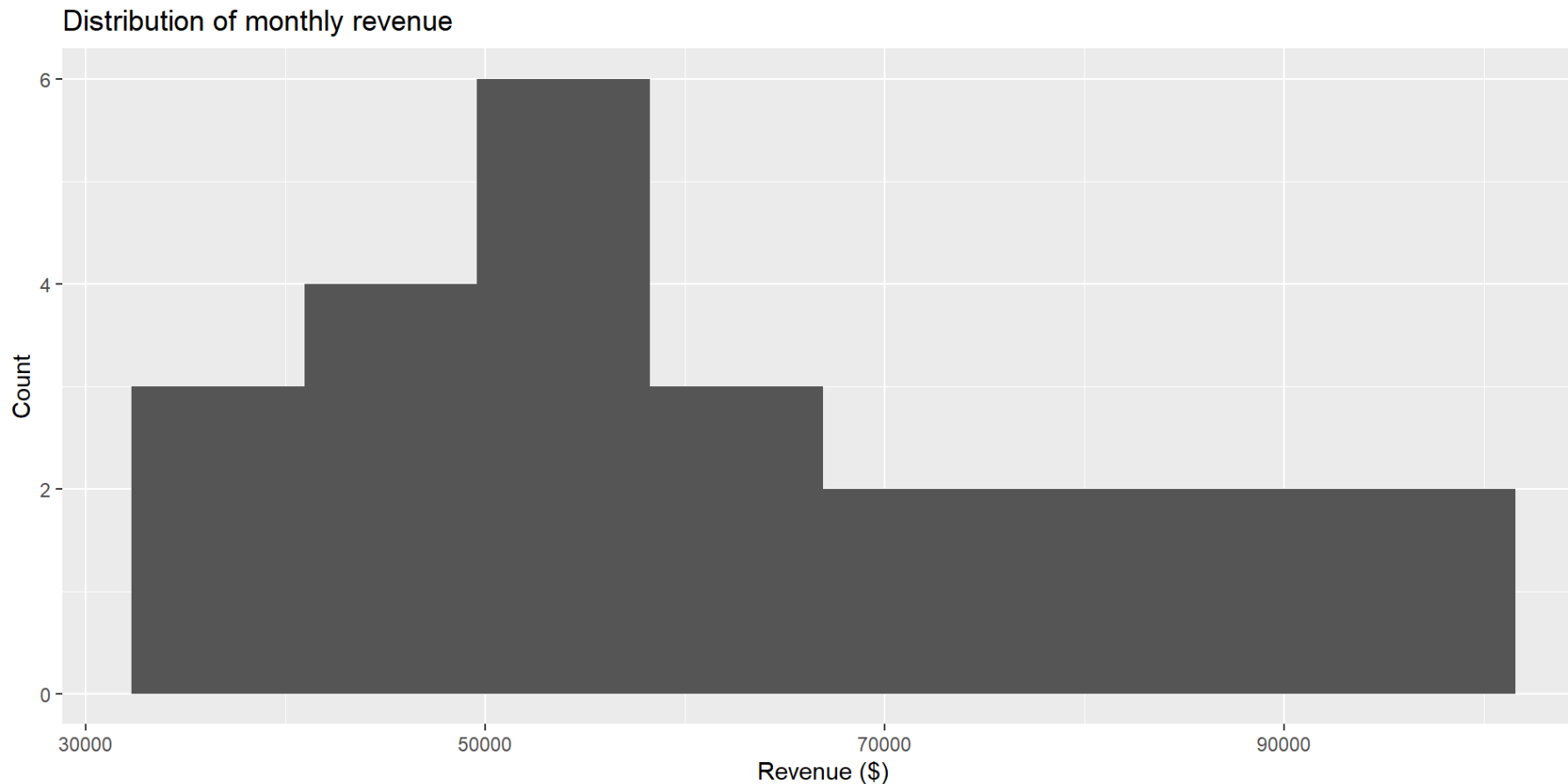
Bar Chart: **region** (Categorical)

```
1 library(tidyverse)
2 sales <- read_csv("store_sales.csv")
3
4 sales |>
5   count(region) |>
6   ggplot(aes(x = region, y = n)) +
7   geom_col() +
8   labs(title = "Store-months by region", x = "Region", y = "Count")
```



Histogram: **revenue** (Quantitative)

```
1 ggplot(sales, aes(x = revenue)) +  
2   geom_histogram(bins = 8) +  
3   labs(title = "Distribution of monthly revenue", x = "Revenue ($)", y = "Count")
```



A Common Mix-Up

Building a bar chart of a **quantitative** variable's raw values (one bar per row) instead of a histogram.

A Richer Table: Grouped Summary

```
1 sales |>
2   group_by(region) |>
3   summarize(mean_revenue = mean(revenue), n = n())
```

A tibble: 3 × 3

	region	mean_revenue	n
	<chr>	<dbl>	<int>
1	Midwest	50516.	8
2	South	72578.	8
3	West	61378.	8

Not just how many — how they *performed*.

Center and Spread

- **Mean** — arithmetic average; sensitive to extreme values
- **Median** — middle value when sorted; resistant to extreme values
- **Standard deviation** — typical distance from the mean

```
1 sales |>
2   summarize(
3     mean_revenue = mean(revenue),
4     median_revenue = median(revenue),
5     sd_revenue = sd(revenue)
6   )
```

```
# A tibble: 1 × 3
```

	mean_revenue	median_revenue	sd_revenue
	<dbl>	<dbl>	<dbl>
1	61491.	57750	17808.

When the Mean Misleads

Add one hypothetical flagship store with unusually high revenue...

- **Mean** shifts noticeably
- **Median** barely moves

Means are pulled toward extreme values; medians are not.

A Mistake Worth Making on Purpose

```
1 mean(sales$region)
```

“Average region” isn’t a real quantity.

A Statistic That Actually Fits

```
1 sales |>  
2   count(region) |>  
3   mutate(proportion = n / sum(n))
```

A tibble: 3 × 3

	region	n	proportion
	<chr>	<int>	<dbl>
1	Midwest	8	0.333
2	South	8	0.333
3	West	8	0.333

“West stores account for about a third of all store-months.”

Never a mean, for a categorical variable. A proportion or a mode instead.

Executive Summary: Strong vs. Weak

Worth **20 of 100 points** — the single largest line item.

Weak (restates a number):

“Average revenue was \$65,000.”

Strong (interprets it, tied to a decision):

“Midwest stores generated noticeably less revenue per month than South or West stores, even after accounting for smaller store size, worth investigating before allocating next quarter’s inventory evenly across regions.”

Your Turn: One Sentence (2 minutes)

Using the grouped table from earlier, write **one sentence** that could open your own executive summary.

Model it on the strong example — not the weak one.

Your Turn: Putting It All Together

Using `store_sales.csv`, individually:

1. **One table** — frequency table or grouped summary
2. **One graph** — a histogram of `units_sold`
3. **Two statistics, matched to type** — mean *and* median of `units_sold` (quantitative), *and* a proportion for one region (categorical)
4. **One sentence** — in the style of the strong executive summary

The Module 1 Deliverable

A short Quarto exploratory report, containing:

- **Framing** (Part 1)
- **Data description** — observational unit + at least 4 variables classified (Part 1)
- **Executive summary**
- At least one **table**
- At least one **graph**
- At least two **descriptive statistics**, correctly matched to variable type

You Already Practiced All of It

- [store_sales.csv](#), or a dataset of your own from your workplace (instructor approval required)
- Every skill above: practiced today, across all three parts